

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- e. Bonus problems are not required for credit, but they are worth extra credit.

Problems

Problem 1. (Gradient Descent for Linear Regression.) Consider the linear regression problem:

$$\min_{\theta} J(\theta) = \min_{\theta} \|X\theta - y\|^2, \quad (1)$$

where $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$. When the problem is overdetermined i.e $n > d$ and X is full column-rank, we saw in Module 2 that $X^\top X$ is invertible and the solution to the problem in (1) is $\hat{\theta}_{\text{OD}} = (X^\top X)^{-1} X^\top y$.

In HW4 and in lecture, we considered the underdetermined case, i.e. $n < d$; in this case, the problem (1) has infinitely many solutions with objective value zero (i.e. such that $\|X\theta - y\| = 0$). If X is full row-rank, then $XX^\top$ is invertible and the minimum norm solution admits the closed form expression $\hat{\theta}_{\text{UD}} = X^\top (XX^\top)^{-1} y$.

In this problem, we will explore the solution found by gradient descent for problem (1) for both cases. Refer to the initial point of gradient descent as θ_0 and the iterates as $\{\theta_t\}_{t=1}^T$. We say that the algorithm converges if there exists a τ such that $\theta_t = \theta_\tau$ for all $t > \tau$, and we refer to $\hat{x}_{\text{GD}} = x_\tau$ as the solution obtained by gradient descent.

- a. **Gradient descent update:** Compute the gradient of $J(\theta)$ and write the gradient descent update

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t)$$

explicitly in terms of data (X, y) , and step-size η .

- b. **Fixed point condition:** Suppose that gradient descent converges to a point θ_∞ . Write down the equation it must satisfy. What is the interpretation of this condition?
- c. **Solution obtained by gradient descent:** Now suppose that we are in the overdetermined case, and assume full column-rank. Write down the solution obtained by gradient descent, assuming that training converges to θ_∞ .

Problem 2. (Gradient Descent for Linear Regression: Underdetermined Case.) Consider a simple special case to develop intuition: let $n = 1, d = 2$, and choose $X = \begin{bmatrix} 2 & 1 \end{bmatrix}$ and $y = [2]$.

- Show that there exist infinitely many $\hat{\theta}$ satisfying $X^\top \hat{\theta} = y$ on a line ℓ . Write the equation of the line ℓ .
- Starting from $\theta_0 = 0$, what is the direction of the negative gradient, written as a unit-norm vector? Does the direction change along the trajectory?
- Now, we explore for the simple example which solution gradient descent finds. Assume that gradient descent converges (as defined in **Problem 1**).
 - Using your findings from the previous part, which solution $\hat{\theta}_{\text{GD}}$ on line ℓ does gradient descent find?
 - Verify that for any other solution $\hat{\theta} \in \ell$, it holds that $(\hat{\theta} - \hat{\theta}_{\text{GD}}) \perp \hat{\theta}_{\text{GD}}$.
 - Through a pictorial sketch, argue that $\hat{\theta}_{\text{GD}}$ has the smallest Euclidean norm across all the solutions on the line ℓ .

Hint: Pythagoras theorem may be helpful here.

Problem 3. (Gradient Descent for Linear Regression: Generalize from Simple Example.) Now, we generalize the argument from the simple example to general underdetermined systems. Assume full row-rank.

- Show that for $\theta_0 = 0$, the iterates are always in the span by the rows of X . (**Hint:** you can use induction to show this).
- Show that $\hat{\theta}_{\text{GD}} = \hat{\theta}_{\text{UD}}$

This result is surprising and nontrivial. It states that amongst infinitely many solutions that attain the optimal objective value, gradient descent chooses the one with the minimum norm, which is a nice desirable property. This phenomenon, known as *implicit regularization*, helps explain the success of using gradient-based methods to train overparameterized models, like deep neural networks; this has been a very active area of research over the last 5 years!

Problem 4. (Step Size Choice in Gradient Descent.)

Consider minimizing a smooth quadratic function

$$f(\theta) = \frac{1}{2} \|X\theta - y\|_2^2$$

where $X \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^m$. Gradient descent updates are given by

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t), \quad \text{with } \nabla f(\theta) = X^\top (X\theta - y).$$

- Write the iteration in matrix form.** Show that the error $e_t = \theta_t - \theta^*$, where θ^* is the least-squares solution, evolves as

$$e_{t+1} = (I - \eta X^\top X) e_t.$$

- Condition for convergence.** Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $X^\top X$. Show that gradient descent converges for any initialization if and only if

$$|1 - \eta \lambda_i| < 1 \quad \text{for all } i.$$

Hence derive the allowable range of step sizes η . Hint: Write $X^\top X = Q\Lambda Q^\top$ with eigenvalues $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then set $e_t = Qz_t$ and show that $z_{t+1} = (I - \eta\Lambda)z_t$.

- (c) **Optimal constant step size.** Within the convergent range, find the value of η that minimizes the worst-case contraction factor

$$\rho(\eta) = \max_i |1 - \eta\lambda_i|.$$

Show that the optimal choice satisfies

$$\eta^* = \frac{2}{\lambda_{\max} + \lambda_{\min}}$$

and compute the corresponding convergence rate ρ^* .

- (d) **Discussion.** Interpret intuitively what happens when 1. η is too small, and 2. η is too large. Relate your observations to the shape (conditioning) of the quadratic bowl $f(\theta)$.